

 SafeSmart
Z-STACK™

The Modular, Pre-Engineered Stair Tower

ATI
ATI-PRODUCTS & SERVICES

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INTRODUCTION



WHO WE ARE

SafeSmart is a global leader in engineered access solutions, trusted by organizations operating in complex, safety-critical environments where reliability, efficiency, and worker protection are paramount. Across construction sites, industrial facilities, aviation maintenance operations, mining projects, energy infrastructure, and large-scale manufacturing environments, SafeSmart systems enable crews to move safely and efficiently between elevations while maintaining productivity and compliance.

For more than three decades, SafeSmart has partnered with contractors, engineers, and facility operators to solve access challenges that conventional ladders, scaffold, and improvised solutions often fail to address. Our portfolio spans a broad range of modular and engineered systems—from portable adjustable stair systems to fully integrated platforms and large-scale access structures—designed to support every stage of a project, from construction and commissioning to inspection, maintenance, and long-term operations.

What sets SafeSmart apart is our commitment to engineering the right solution for each environment. Every system is developed with a deep understanding of real-world site conditions, operational demands, and safety requirements. Through advanced modular design, high-strength materials, and practical field experience, SafeSmart delivers access systems that reduce risk, improve workflow, and eliminate inefficient methods of working at height.

Safe, reliable access is not optional in today's industries—it is fundamental to keeping projects moving and people protected. That is why organizations around the world rely on SafeSmart as the trusted authority in engineered access solutions.

TRUSTED BY THE PROS



Georgia-Pacific

BAE SYSTEMS



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Introducing **Z-STACK™**

Z-Stack™ is SafeSmart's engineered stair tower system designed to provide safe, efficient vertical access on complex worksites where traditional solutions fall short. Built as a modular, freestanding structure, Z-Stack allows crews to move safely between multiple elevations while maintaining consistent stair geometry, proper guardrail protection, and predictable worker flow across the structure.

The system is based on a bolt-together, stackable design that allows stair and landing modules to be combined to reach the required height while maintaining stability and structural integrity. As projects evolve, the tower can be expanded, repositioned, or reconfigured to meet changing site conditions without the ongoing adjustments and inefficiencies often associated with traditional scaffold stair towers.

Developed to address the access challenges of large construction, infrastructure, and industrial projects, Z-Stack™ has become a trusted solution for contractors who require reliable, repeatable vertical access. Its combination of engineered modular components, rapid installation, and long-term durability makes Z-Stack™ a practical alternative to conventional scaffold stair systems and an essential tool for modern worksites.

Summary of Benefits

- **A Smarter Alternative to Scaffold Stair Towers** – provides safe, engineered vertical access between multiple elevations without the inefficiencies of traditional scaffold structures.
- **Rapid Modular Assembly** – bolt-together components allow stair towers to be erected quickly with predictable installation sequences.
- **Scalable Height Configurations** – stackable modules allow the system to be configured for varying elevations across changing worksites.
- **Consistent Stair Geometry** – integrated stairs and landings maintain proper step spacing, guardrail protection, and safe worker movement throughout the tower.



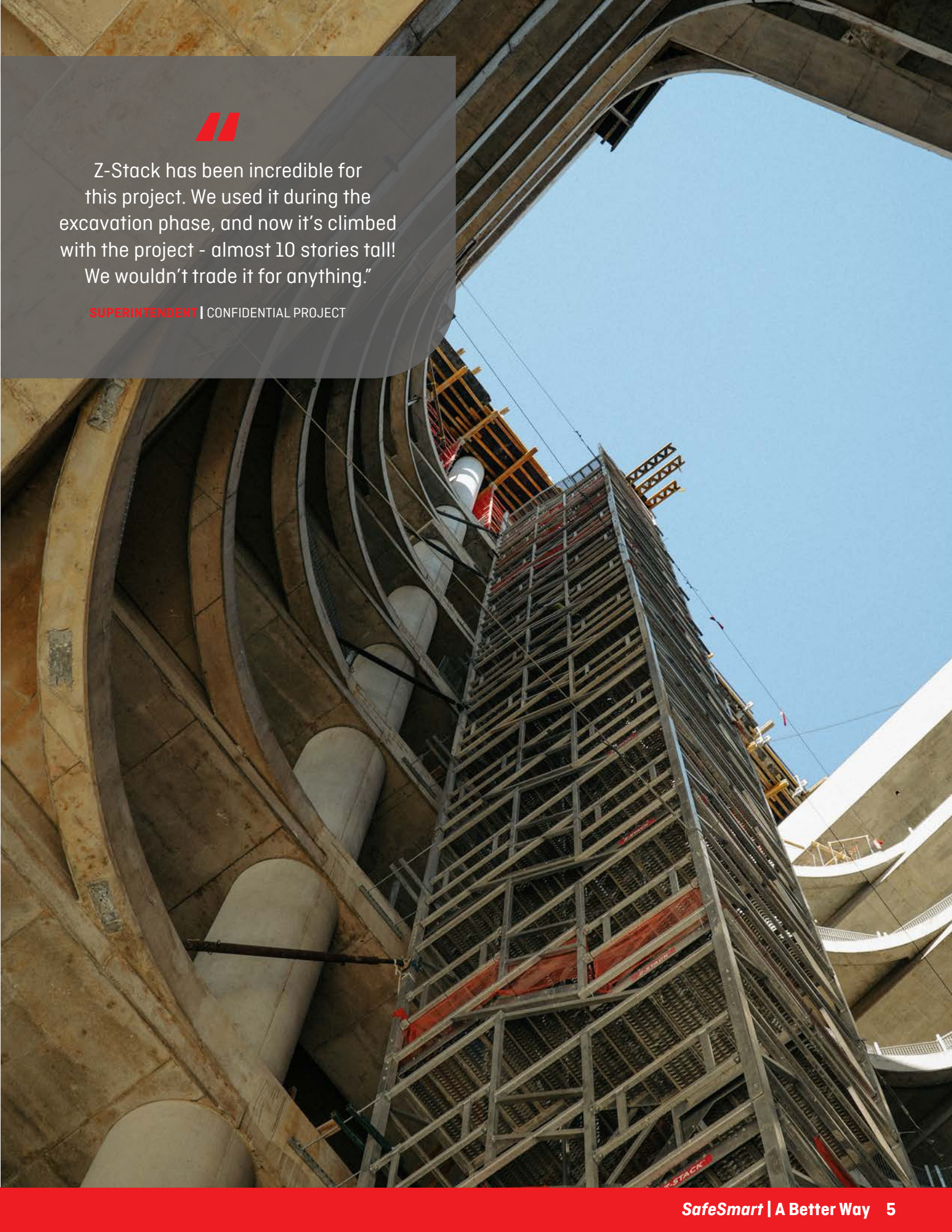
Z-STACK™ is a patented SafeSmart innovation.

Developed and engineered by SafeSmart, Z-Stack™ represents an advanced modular stair tower solution designed to provide safe, scalable vertical access on demanding worksites. The Z-Stack™ name, system design, and associated technologies are proprietary to SafeSmart and protected under applicable intellectual property protections.



Z-Stack has been incredible for this project. We used it during the excavation phase, and now it's climbed with the project - almost 10 stories tall! We wouldn't trade it for anything."

SUPERINTENDENT | CONFIDENTIAL PROJECT



The Problem with Scaffold & Ladders

Traditional Access Gets the Job Done. Z-Stack™ Gets It Done Better.

Across construction sites, industrial facilities, infrastructure projects, and maintenance environments, scaffold stair towers, ladders, and cage ladders are commonly used to access elevated work areas. Yet these solutions often introduce limitations that impact both worker safety and operational efficiency, particularly when they require certified erection crews, frequent inspections, and labor-intensive assembly using heavy steel components.

When crews must rely on ladders or assemble scaffold structures to move repeatedly between elevations, setup time increases, worker flow slows, and safety risks can grow. In permanent installations, cage ladders and fixed vertical access can also create difficult and impractical routes for workers carrying tools or equipment, turning what should be simple access into a daily operational challenge.

Where Scaffold & Ladders Fall Short

Labor-Intensive - Scaffold stair towers require significant labor and time to assemble before crews can safely access elevated work areas.

Restricted Worker Movement - Ladders and cage ladders limit worker flow and make it difficult to safely carry tools or equipment between levels.

Inefficient Temporary Access - Traditional scaffold systems are often overbuilt for short-term access yet still require ongoing adjustment as site conditions change.

Poor Ergonomics for Daily Use - Vertical ladders and cage ladders create physically demanding access routes that slow productivity and increase fatigue.



COMPLIANCE & SAFETY NOTE

Did You Know?

OSHA requires scaffold systems to be inspected by a competent person before each work shift and after any event that could affect structural integrity.

Z-Stack™ reduces this burden with a rigid, engineered stair tower system designed for stable, repeatable access without the constant inspection and adjustment required by scaffold.

Source: OSHA 29 CFR 1926.451(f)(3)



Why Choose **Z-STACK™**

The Craneable, Pre-Engineered Stair Tower

Z-Stack was engineered to provide a safer, more efficient alternative to traditional scaffold access, ladders, and cage ladders. With its modular stair tower design, fully bolted locknut-secured connections, integrated landings, and durable aluminum construction, Z-Stack creates a rigid, craneable access system that allows crews to move safely between elevations while carrying tools and equipment, delivering consistent, repeatable access on complex worksites.

Benefit	Description
Modular Vertical Access	Configurable tower modules allow Z-Stack to be built to multiple heights and layouts, delivering safe, repeatable access between elevations without custom fabrication.
Faster Installation Than Scaffold	Pre-engineered components bolt together quickly, allowing crews to install, and reposition access towers significantly faster than traditional scaffold systems.
Safer Worker Movement	Integrated stairs, guardrails, and landings allow workers to move safely between levels while carrying tools and equipment.
Reusable Across Projects	Durable aluminum construction allows Z-Stack systems to be dismantled, relocated, and reused across multiple jobsites.
Built for Demanding Environments	High-strength aluminum construction and compliant guardrail systems provide stable, reliable access in construction and industrial environments.



The Smarter Alternative to Ladders and Scaffold

Traditional ladders and scaffold were never designed for the speed and demands of modern worksites. Z-Stack delivers a safer, faster, and more efficient way for crews to move between elevations and reposition as needed on-site.

See how Z-Stack compares to traditional access methods.

CAPABILITIES	Z-STACK™	SCAFFOLD	LADDERS
Safe worker access between levels	✓	✓	✗
Carry tools & equipment safely	✓	✓	✗
Multiple workers simultaneously	✓	✓	✗
Rapid setup (hours vs. days)	✓	✗	✓
Reusable across job sites	✓	✗	✓
Modular, scalable height system	✓	✓	✗
Continuously compliant design	✓	✗	✗
Stable stair access vs vertical climb	✓	✓	✗
Improves worker flow between levels	✓	✗	✗
Craneable & easy to relocate	✓	✗	✓

Engineered Performance

Key Benefits of the Z-STACK™ System

Z-Stack is purpose-built with precision engineering and high-tensile aluminum construction to deliver exceptional strength without unnecessary weight. The system's structural design provides the rigidity required for demanding worksites while remaining lightweight enough to be efficiently positioned, repositioned, and redeployed as site conditions evolve. This balance of strength, durability, and mobility ensures Z-Stack performs reliably across both temporary and permanent access applications.



Craneable

Designed with integrated lifting points for safe crane placement during installation.



Lightweight Aluminum

High-strength aluminum construction reduces weight while maintaining structural durability.



Fast Installation

Pre-engineered bolt-on components allow for quick, efficient assembly on site.



Reusable & Permanent Use

Suitable for reusable temporary deployment and long-term installed access solutions.



Simplified Transport

Flat-packable components simplify transport, storage, and jobsite logistics.

A Closer Look

Innovation in Every Detail

Fully Bolted Connection

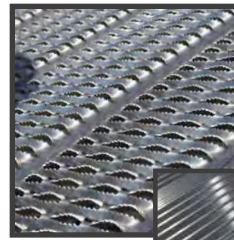
Fully bolted, locknut-secured connections allow the tower to be safely crane-lifted and relocated, reducing downtime and avoiding daily scaffold inspections.



MODULE JOINER & CRANE HOOK

Modules connect with integrated joiners that also serve as crane hooks for lifting individual modules.

Lifting Bars available for lifting fully assembled structure. [See Page 12]



ANTI-SLIP TREADS

Provides secure footing in demanding environments. Grip-Strut Treads available.



TOEBOARDS

Integrated toeboards can be added before or after assembly, preventing tools, debris, and other materials from falling from the tower.



FULLY WELDED FRAMES

Pre-fabricated frames simplify installation by minimizing components, while increasing overall system strength. [See Page 14]



HEIGHT-ADJUSTABLE BASE JACKS

Anchors the tower to approved surfaces, and adjusts height to level the system on uneven ground.

Castors & Mobile Chassis customizations available. [See Page 13]

Standard Option

Full-Size Tower Configuration

Standard Z-Stack towers offer a wider access system suited for construction and industrial environments, supporting higher worker traffic and the movement of tools and materials between levels.

Access Width - 36"



Common Use Cases

Construction Site Access | Emergency Egress Towers | Large Crew Access | Multi-Level Construction Access | Industrial Roof Access | Infrastructure Projects

TOP STANDARD MODULE

DIMENSIONS	
LENGTH	WIDTH
9'-4"	6'-10"

COMPONENTS	
QTY	DESCRIPTION
1	Right Frame
1	Left Frame w/ (1) Gate
1	Brace Frame
1	Brace Frame w/ (2) Gates
1	Middle Handrail
1	Top Handrail
4	Vertical Connectors (for Craning)

MID-LEVEL STANDARD MODULE

DIMENSIONS	
LENGTH	WIDTH
9'-4"	6'-10"

COMPONENTS	
QTY	DESCRIPTION
1	Right Frame
1	Left Frame w/ (1) Gate
1	Brace Frame
1	Brace Frame w/ (2) Gates
1	Middle Handrail
2	Z-Stack Stairs w/ 4-Plank Deck
4	Vertical Connectors

BOTTOM STANDARD MODULE

DIMENSIONS	
LENGTH	WIDTH
9'-4"	6'-10"

COMPONENTS	
QTY	DESCRIPTION
1	Right Frame
1	Left Frame w/ (1) Gate
1	Brace Frame
1	Middle Handrail
2	Z-Stack Stairs w/ 4-Plank Deck
4	Vertical Connectors
4	Base Jack Holder
4	Adjustable Base Jacks

Compact Option

Reduced Footprint Configuration

Compact Z-Stack towers provide safe vertical access with a reduced footprint, serving as a true ladder replacement for roof access, equipment platforms, and permanent facility access where space is limited.

Access Width - 24"



Common Use Cases

Roof Access | Ladder Replacement | Mechanical Equipment Access | Utility Platform Access | Facility Service Access | Confined Site Installations | Plant & Facility Access

TOP COMPACT MODULE

DIMENSIONS

LENGTH	WIDTH
7'-7"	4'-11"

COMPONENTS

QTY	DESCRIPTION
1	End Frame
1	Access End Frame
1	Left Frame
1	Right Frame
1	Middle Handrail
1	Top Handrail
4	Vertical Connectors (for Craning)

MID-LEVEL COMPACT MODULE

DIMENSIONS

LENGTH	WIDTH
7'-7"	4'-11"

COMPONENTS

QTY	DESCRIPTION
1	End Frame
1	Access End Frame
1	Left Frame
1	Right Frame
1	Middle Handrail
2	Compact Z-Stairs 24" Wide
4	Vertical Connectors

BOTTOM COMPACT MODULE

DIMENSIONS

LENGTH	WIDTH
7'-7"	4'-11"

COMPONENTS

QTY	DESCRIPTION
1	End Frame
1	Access End Frame
1	Left Frame
1	Right Frame
1	Middle Handrail
2	Compact Z-Stairs 24" Wide
4	Vertical Connectors
4	Z-Stack Adjustable Base Jack Kits

Accessories & Attachments

Z-Stack can be enhanced with a range of engineered accessories designed to expand functionality, improve site integration, and adapt the system to specific project requirements. These components allow the tower to be configured for different environments, access points, and operational needs while maintaining the safety, stability, and modular flexibility that define the Z-Stack system.

Lifting Bars

Multi-Module Lifting System

Lifting bars allow the Z-Stack tower to be safely crane-lifted as a complete structure while maintaining rigidity and preventing twisting during lifting.

Key Benefits:

- Enables crane-assisted lifting and repositioning
- Distributes lifting loads across modules
- Prevents tower twisting during lifting
- Ideal for repositioning towers on large sites



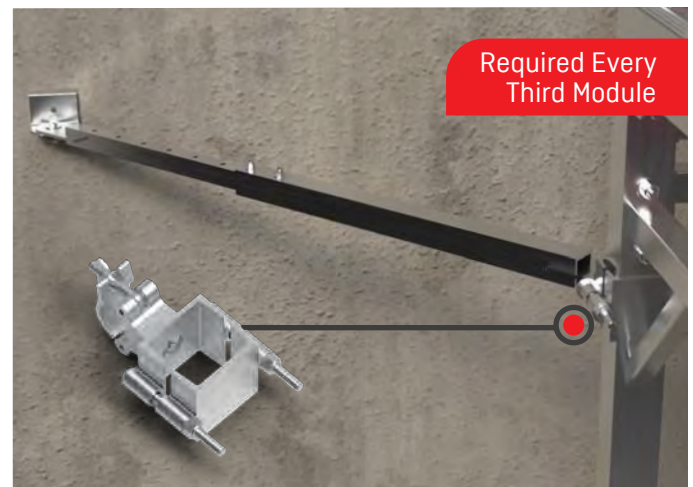
Tie-Off Brackets

Structural Stabilization System

Telescopic Tie-Off Brackets secure the Z-Stack tower to a nearby structure, providing added stability and reducing sway.

Key Benefits:

- Secures tower to adjacent structures
- Reduces sway at higher elevations
- Required on every third module for compliance
- Simple bolt-on installation



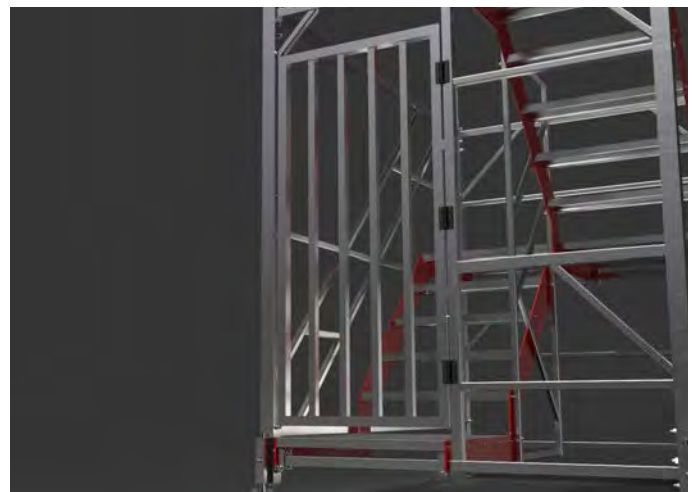
Entry Gates

Controlled Tower Access

Entry gates install at the base module to control tower entry and prevent unauthorized access to the stair system. Mesh-covered entry gates are available.

Key Benefits:

- Prevents unauthorized tower access
- Attaches directly to entry frame
- Available in aluminum or mesh designs



Security Mesh Options

Fall & Debris Protection

Security Mesh Panels can be installed along tower sections to help contain debris and limit unauthorized climbing.

Key Benefits:

- Helps prevent falling tools and debris
- Restricts unauthorized access
- Multiple mesh options available
- Improves jobsite safety



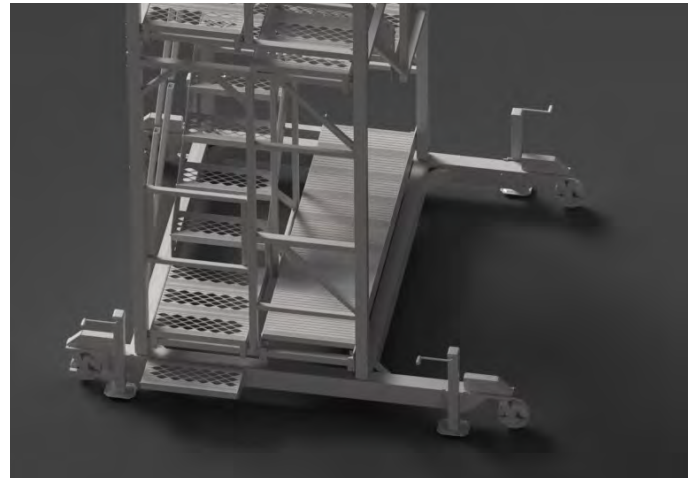
Mobile Chassis System

Mobile Tower Base

The mobile chassis system allows Z-Stack towers to be repositioned within controlled environments using heavy-duty lockable castors.

Key Benefits:

- Allows towers to be moved easily
- Ideal for indoor industrial environments
- Locking castors provide stability
- Optional castor systems for low-rise towers



Removable Cantilever System

Extended Access Platform

The removable cantilever system extends the reach of the tower to provide access to structures or equipment beyond the tower footprint.

Key Benefits:

- Extends working reach from tower
- Ideal for maintenance access at height
- Modular and removable design
- Compatible with multiple tower setups



NEED CUSTOM OPTIONS? CALL TODAY! 1.800.869.1525

SafeSmart has provided custom solutions to numerous industries including: Aviation, Mining, Aerospace, Industrial, Construction, & more.

How Bolt-On Parts Assemble the Z-Stack™

3-Module Types - Endless Heights.

Pre-engineered components bolt together using integrated connection points and supplied hardware, allowing each module to be assembled quickly and accurately on site. Middle modules can then be repeated as needed to extend the tower to the desired height, creating a rigid, durable structure.

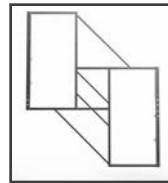
Top Module (x1)

PART DESCRIPTION	QTY
Top Handrail	1
Diagonal Brace	1
Left Brace	1
Entry Frame	1
End Frame	1
Gate Stand	1

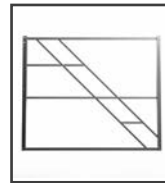
Top Handrail



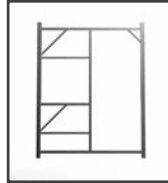
Diagonal Brace



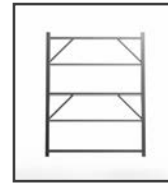
Left Brace



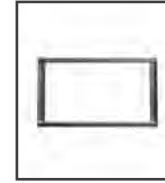
Entry Frame



End Frame



Gate Stand



Optional Joiner & Lift Bars

The top module can be fitted with additional joiners and a lifting bar to allow the entire tower to be safely crane-lifted as one structure.

See Page 12 for more information.

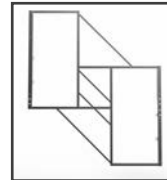
Middle Modules (x Endless)

PART DESCRIPTION	QTY
End Frame	1
Diagonal Brace	1
Left Brace	1
Middle Handrail	1
Gate Stand	1
Joiner	4
Z-Stairs	2

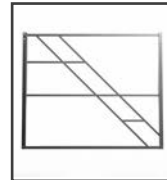
End Frame



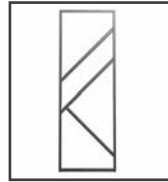
Diagonal Brace



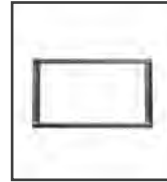
Left Brace



Middle Handrail



Gate Stand



Joiner



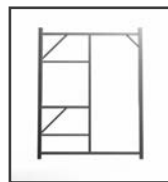
Z-Stairs



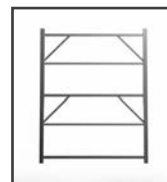
Base Module (x1)

PART DESCRIPTION	QTY
Entry Frame	1
Left Brace	1
End Frame	1
Z-Stairs	2
Middle Handrail	1
Base Jack	4
Joiner	4

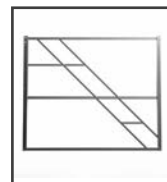
Entry Frame



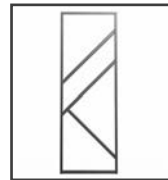
End Frame



Left Brace



Middle Handrail



Base Jack



Joiner



Z-Stairs



How Z-Stack™ Modules Stack

The Tower That Grows With The Job

Scan to View the
Z-Stack™
Webpage

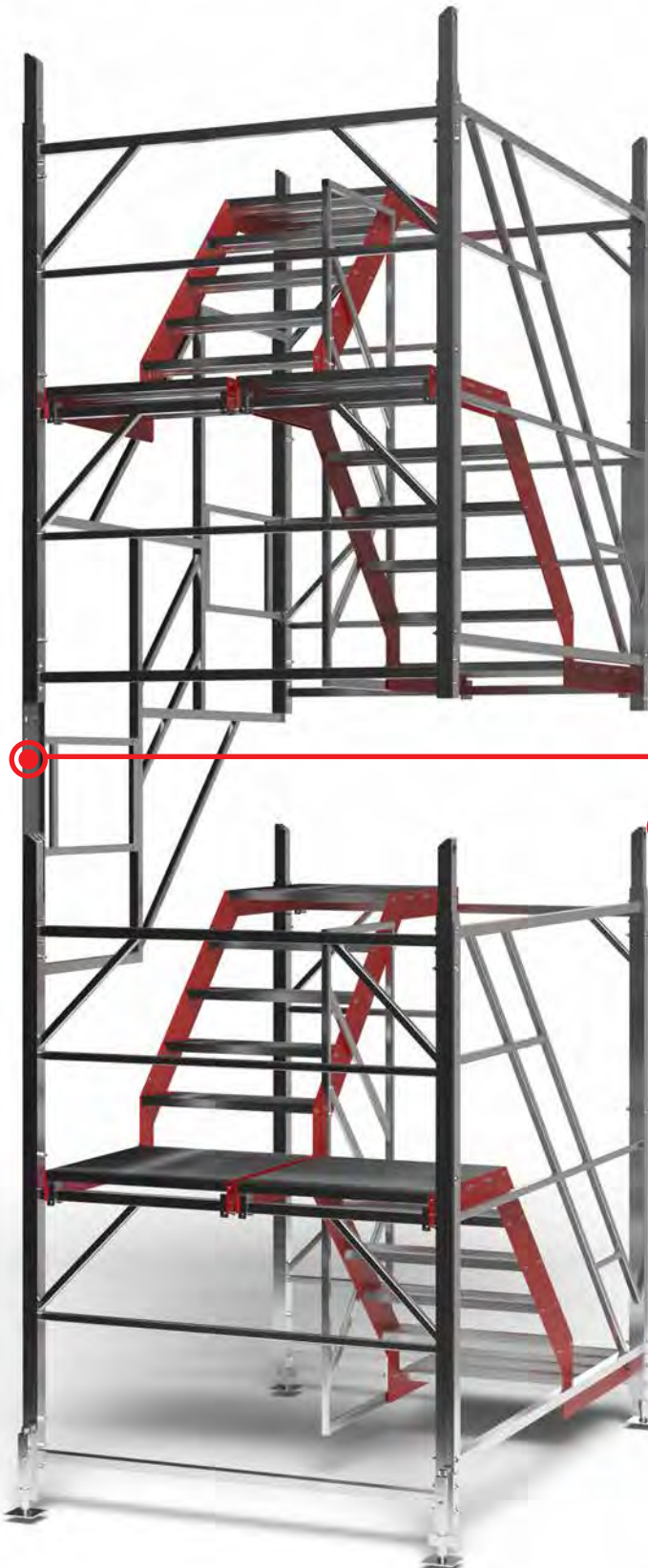


Male-to-Female Joiner System

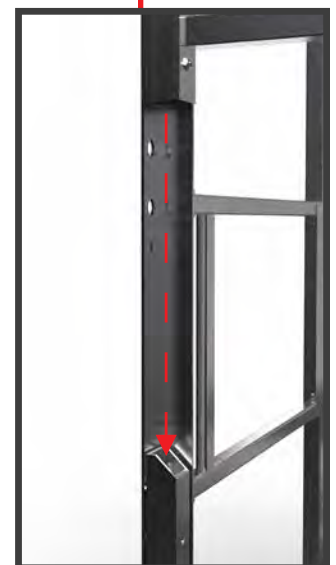
Plug-and-Play Modular Connection

Z-Stack modules are designed with an integrated male-to-female joiner system that allows towers to be stacked quickly and securely. Each module is lifted into place and lowered onto the four structural joiners extending from the module below. Once seated, the connection points are secured using the included bolts, locking the modules together into a rigid tower structure.

Modules with staggered wall sections connect in the same manner. The extended wall section drops into the level below and is secured using the integrated L-bracket connection, reinforcing the frame and maintaining structural alignment as the tower grows.



Standard Joiners (x4)



L-Shaped Bracket Wall

Safe Landing Solutions

Ensuring Continuous Safe Access at the Exit Point

Because Z-Stack towers are modular, the final landing height may not always perfectly match the elevation of the structure being accessed. SafeSmart provides engineered transition solutions that allow workers to safely step on or off the tower without unsafe gaps or improvised site-built solutions.

Depending on the height difference, systems such as AdaptaStep™ bridging platforms or AdjustaStairs™ can be installed to create a safe, compliant transition—ensuring continuous protected access from ground level to the final destination.

9.6" to 24" Height Difference Solution

ADAPTASTEP™ Bridging

AdaptaStep provides a modular bridging platform designed to safely span elevation differences between the Z-Stack™ landing and the destination surface, while seamlessly integrating into AdaptaStep catwalks.

Key Benefits:

- Quick bolt-on installation to Z-Stack
- Adjustable modular configuration
- **Spans up to 30ft long!**
- Ideal for rooftop, platform, and equipment access



24" to 48" Height Difference Solution

ADJUSTASTAIRS™

AdjustaStairs provides a self-leveling stair transition that safely connects the Z-Stack landing to higher or lower elevations.

Key Benefits:

- Self-leveling treads adjust to height differences
- Rapid installation and repositioning
- Maintains safe stair access between levels
- Eliminates unsafe step-downs or improvised stairs



Engineered Access from Ground to Destination

SafeSmart solutions ensure workers remain protected through every stage of vertical access—from ground level, up the Z-Stack tower, and safely onto the final structure.



Z-Stack helped Nordic PCL overcome some very challenging access issues on our project in Waikiki. SafeSmart is really doing some next-level stuff with this system.

HSE MANAGER | NORDIC PCL



Industry Applications

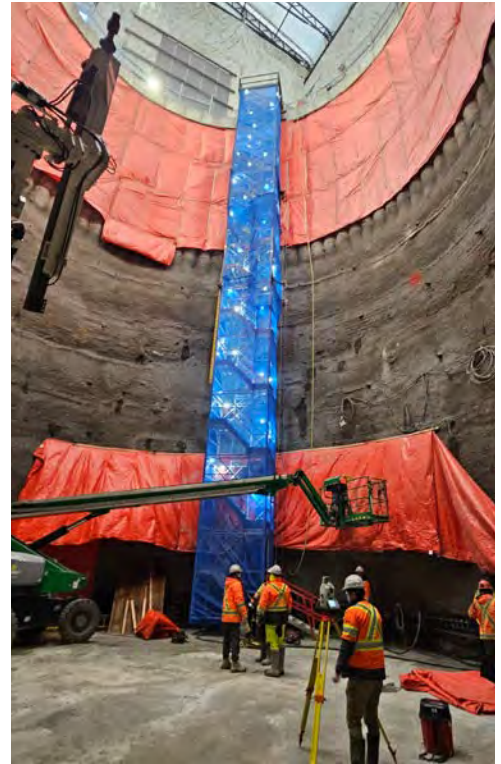
Innovative Access for All Industries

Excavation Access

Safer Access in Deep Excavations

Z-Stack provides stable vertical access into deep excavations where scaffold systems are difficult to install and maintain.

- Modular towers extend safely into deep excavations
- Stair access improves worker movement and safety
- Landings break up long vertical climbs
- Ideal for utilities, infrastructure, and foundation work

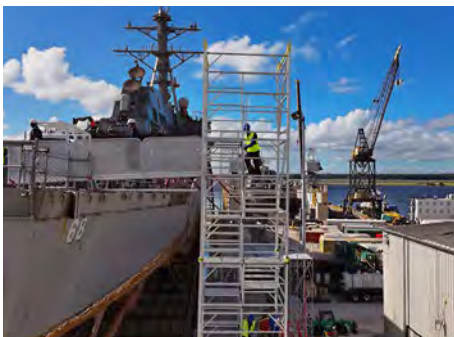


Maritime Dry Dock Access

Crucial Ship Maintenance Access

Z-Stack delivers safe, modular access between dock levels and vessels during maintenance, inspection, and shipyard operations.

- Modular towers adapt to varying dock heights
- Provides safe stair access between work levels
- Durable aluminum resists marine environments
- Ideal for shipyards, dry docks, and vessel servicing



Industry Applications

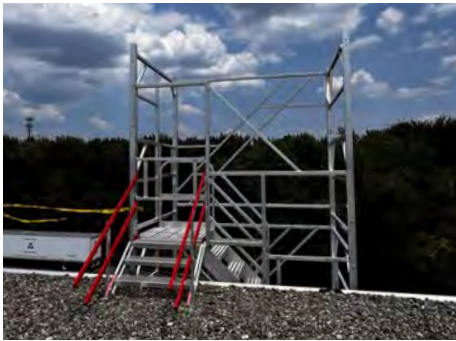
Innovative Access for All Industries

Facility Roof Access

Maintenance & Operations Access

Z-Stack provides a permanent stair tower solution for safe, compliant access to rooftop equipment and mechanical systems.

- Provides safe stair access to rooftop equipment
- Reduces reliance on ladders and cage ladders
- Can be enclosed for controlled facility access
- Ideal for mechanical systems and rooftop service

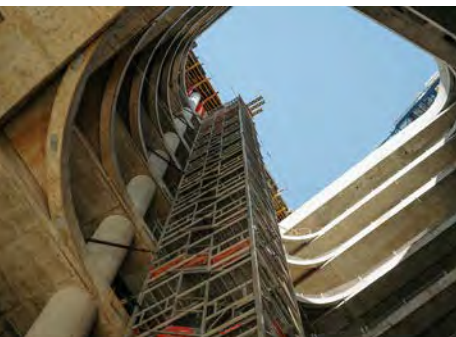


Multi-Level Construction

Access Between Multiple Construction Levels

Z-Stack creates safe stair access between floors on active construction sites, replacing scaffold stair towers and improving worker movement.

- Modular towers scale as construction progresses
- Can be repositioned without deconstruction or rebuild
- Allows safe transport of tools and materials
- Ideal for multi-story construction projects



Project Profile: INNOVATION IN THE HEART OF WAIKIKI



On a high-rise construction site in the heart of Waikiki, SafeSmart partnered with PCL Construction to take access innovation to new heights—literally. This project marked the first-ever deployment of a Z-Stack system fitted with a custom-engineered steel rostrum, purpose-built for this specific structure. The platform projected off the face of the building and carried the entire Z-Stack stair tower without a single anchor point, relying instead on adjustable props that transferred the load into the building's concrete slabs. A bold and self-supporting approach, it brought safety, speed, and structural creativity to one of the most complex sites in the region.



Key Challenges

This project demanded a flexible, non-invasive solution to overcome several site limitations.

- No ability to anchor into building structure
- Tight site footprint in busy downtown location
- Last-minute design changes to the rostrum prior to install
- On-site crew unfamiliar with SafeSmart systems

Z-STACK™ Advantages

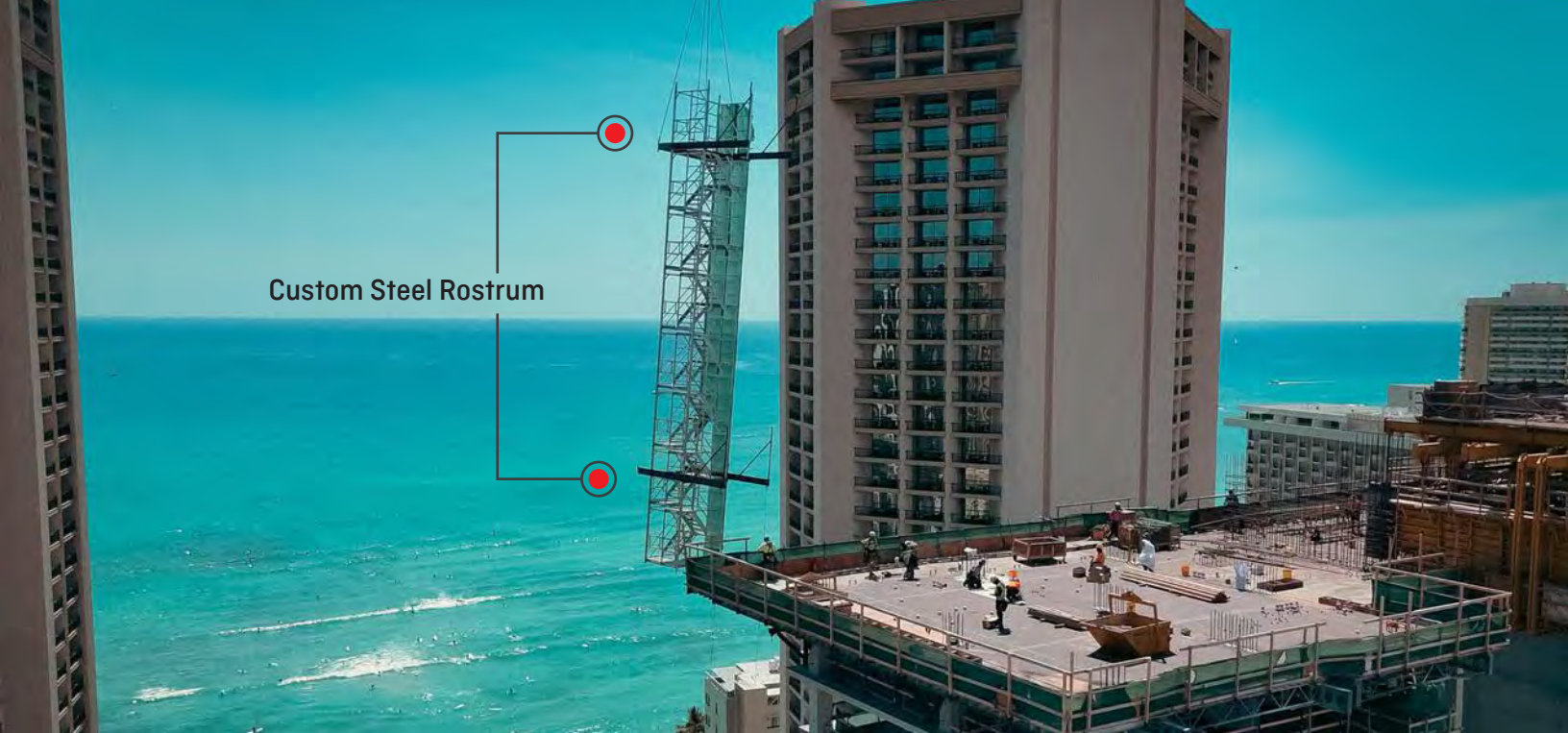
Z-Stack's modular design and pre-planned execution enabled the team to overcome every challenge.

- Rostrum transferred tower load to slab via adjustable props
- No anchors or penetration into structure required
- Fully assembled stair tower lifted into place in one hoist
- Intuitive setup allowed first-time users to assemble with ease
- Maintained safety & efficiency with no impact to structural integrity

The Outcome

The Waikiki project is a showcase of what's possible with Z-Stack—an innovative solution built for the pace and precision of modern construction. From tight space constraints to structural limitations, every challenge was solved with smart engineering and seamless execution. This wasn't just an access system—it was a smarter way to build.

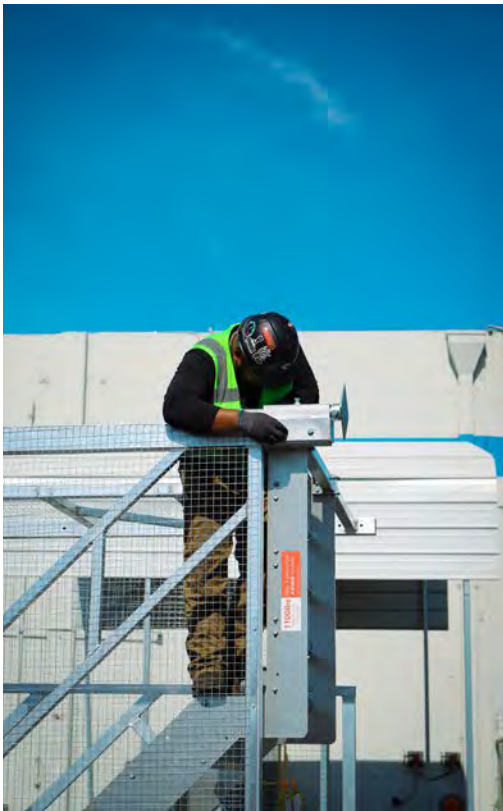




Project Profile: CUSTOM ROOF ACCESS MADE EASY



Georgia-Pacific partnered with SafeSmart to implement a permanent rooftop access solution designed for safe, reliable equipment maintenance. SafeSmart delivered a fully enclosed Z-Stack tower customized to meet the site's operational requirements. Pre-assembled at our local facility, the tower arrived ready for installation. Wrapped in durable steel mesh to provide security from unauthorized access, while maintaining visibility and airflow, the system was engineered as a long-term access solution—eliminating the need for ladders or temporary workarounds.



Key Challenges

This high-traffic facility required a safe, permanent solution that could be installed efficiently.

- Outdated ladder access posed safety and compliance risks
- Uneven ground at install site required adjustable base
- Project needed to be completed with minimal disruption
- Structure had to be fully enclosed for fall protection

Z-STACK™ Advantages

Z-Stack provided a fast, stable, and compliant access solution tailored for industrial use.

- Delivered pre-assembled for fast deployment
- Adjustable feet ensured a level, secure foundation
- Steel mesh enclosure provided full fall protection
- Entire 5-module tower lifted and set in one clean motion
- Bridging walkway connected tower directly to roof access point

The Outcome

The completed Z-Stack system now serves as a permanent, code-compliant rooftop access point—eliminating the need for ladders and unsafe workarounds. The Georgia Pacific project demonstrates how Z-Stack can transform rooftop safety with long-term, maintenance-friendly access built for industrial demands.

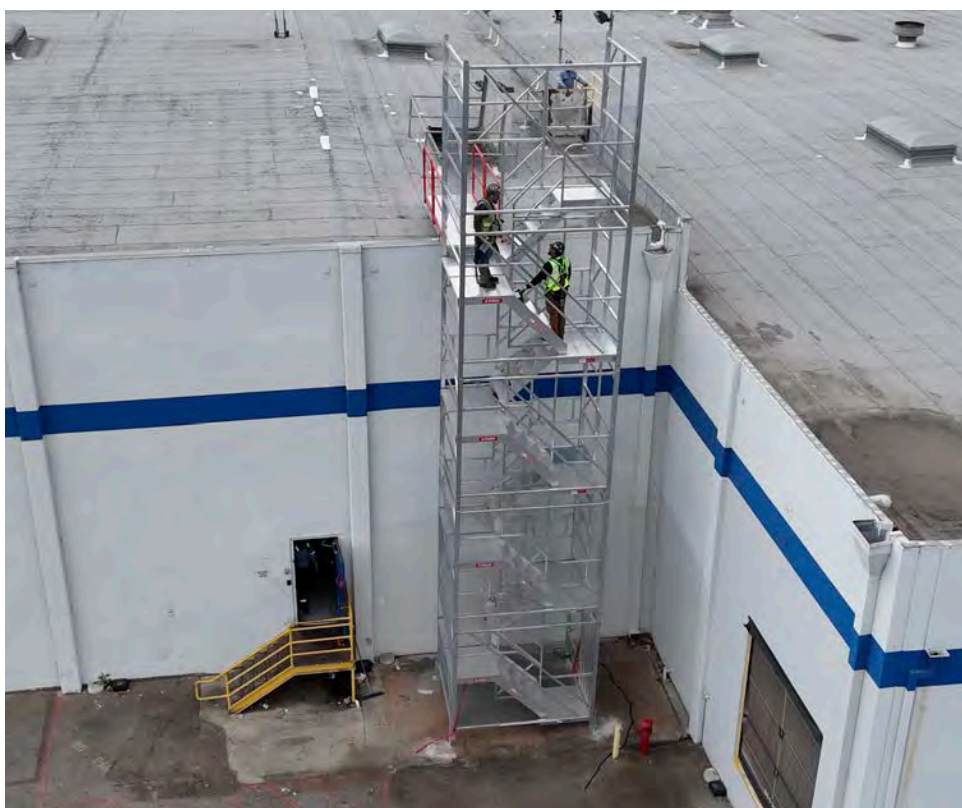
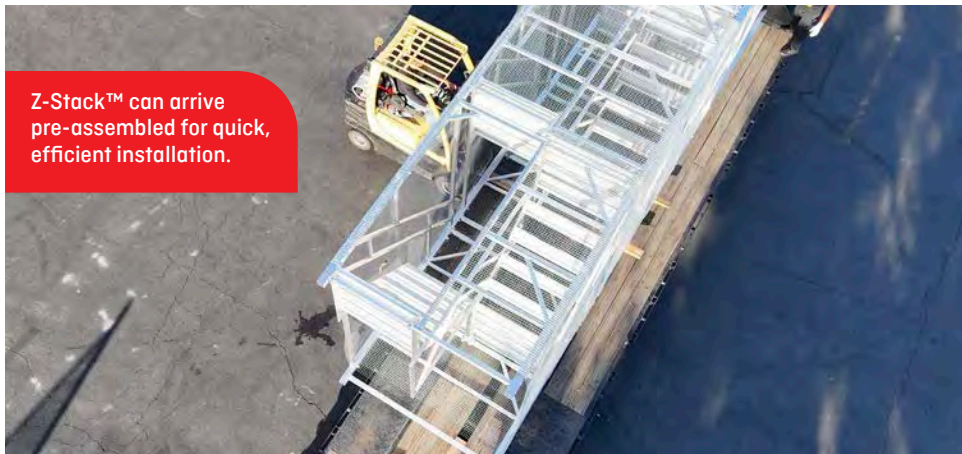




Wire Mesh Exterior
for added security.
See Page 13



Z-Stack™ can arrive
pre-assembled for quick,
efficient installation.



A Better Way

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